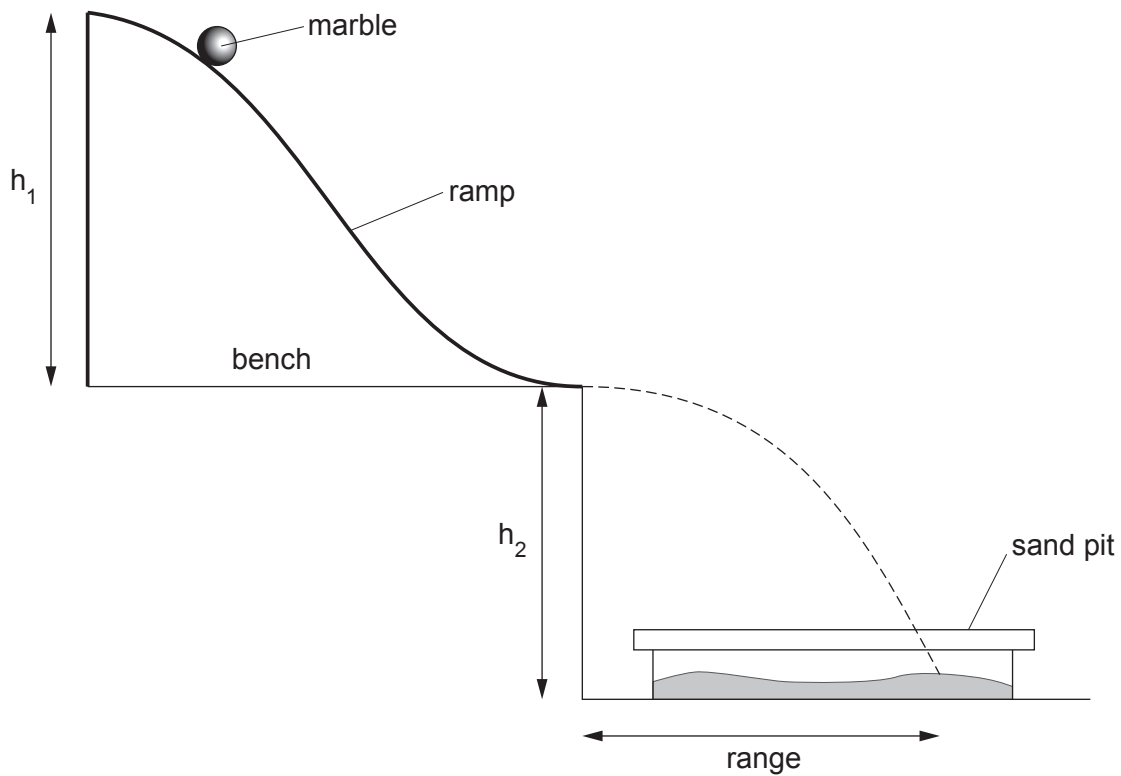


4. A model ski jump can be set up on a bench as shown in the diagram below.



The marble is released from rest at the top of a ramp. It leaves the bottom of the ramp with a certain velocity. It moves through the air and lands in a sand pit. The horizontal distance travelled in the air is called the **range**.

The velocity of the marble as it leaves the ramp is calculated using the equation:

$$\text{velocity} = \frac{\text{range}}{\text{time to travel the range}}$$

The acceleration of the marble down the ramp is calculated using the equation:

$$\text{acceleration} = \frac{\text{velocity of marble as it leaves the ramp}}{\text{time to travel down the ramp}}$$

- (a) The following measurements were recorded during one trial of the experiment.

Range = 45 cm

Time to travel the range = 0.5 s

Time to travel down the ramp = 1.2 s

Use the equations and information above to calculate the acceleration of the marble down the ramp in  $\text{m/s}^2$ . [4]

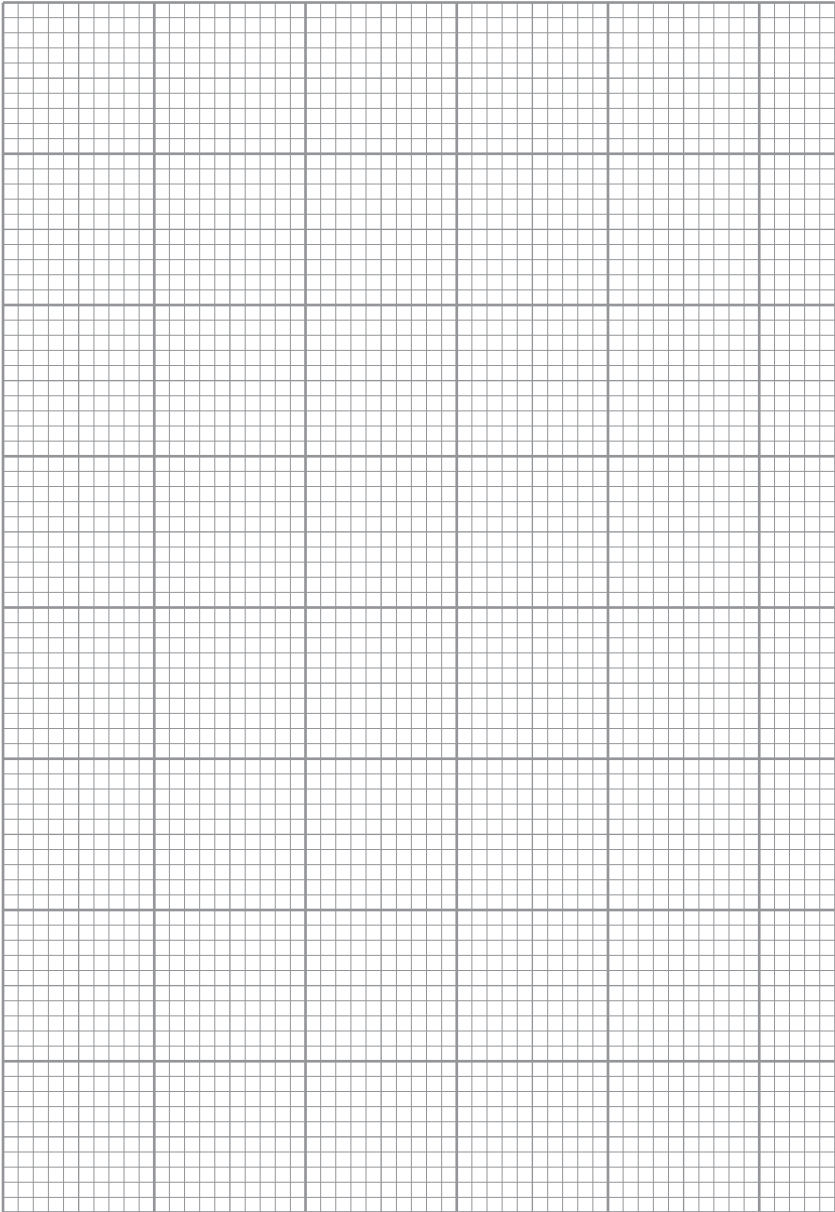
Acceleration = .....  $\text{m/s}^2$

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only

- (b) An experiment is then carried out to find out how the acceleration of the marble depends on the height  $h_1$ . Throughout this experiment height  $h_2$  is constant. The results of the experiment are shown in the table below.

| Height $h_1$ (m) | Acceleration ( $\text{m/s}^2$ ) |
|------------------|---------------------------------|
| 0                | 0                               |
| 0.2              | 0.14                            |
| 0.5              | 0.36                            |
| 0.6              | 0.43                            |
| 0.9              | 0.62                            |
| 1.0              | 0.74                            |

- (i) Use the data in the table to plot a graph on the grid below and draw a suitable line. [3]



- (ii) Use the graph to find the height required to produce an acceleration of  $0.5 \text{ m/s}^2$ . [1]

Height = .....

- (iii) Describe how the experiment can be extended to determine how the range depends on the height  $h_2$ . [2]

.....

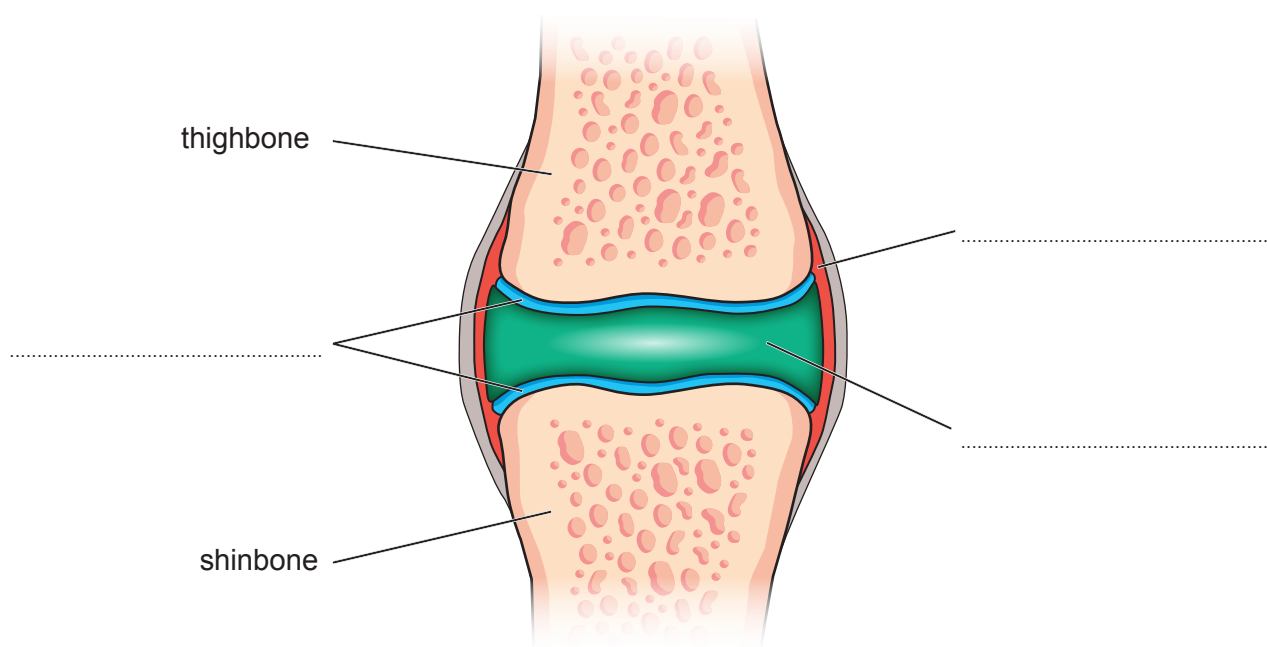
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- (c) Ski jumpers place their knee joints under considerable strain during their career. The knee is easily injured.

A simplified diagram of a knee joint is shown below.



- (i) **Complete** the labelling of the diagram above. [3]
- (ii) Explain why a ski jumper will not be able to compete if they damage the cartilage in their knee joints. [2]

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.....

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- (iii) An X-ray was taken of a ski jumper's leg following an accident.



State the name of this **type** of fracture. ....

[1]

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